

BOW ISLAND, AB, Canada

WMO: 712310

Lat: **49.7344N**Lon: **111.4503W**Elev: **2679**StdP: **13.33**Time Zone: **-7.00 (NAM)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.8	-14.3	-27.2	1.3	-20.6	-20.4	2.0	-14.0	33.5	33.0	29.9	31.8	7.0	160	0.674

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.9	89.3	64.4	85.9	63.1	82.5	62.3	68.2	83.0	66.1	80.6	64.2	78.1	8.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.2	95.8	74.9	60.9	88.1	72.7	58.8	81.7	70.8	34.1	82.8	32.4	80.7	30.8	77.9	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.1	24.4	21.2	-28.7	95.2	6.5	2.6	-33.3	97.1	-37.1	98.7	-40.7	100.2	-45.4	102.1		
(5)			-29.3	73.3	7.6	2.5	-34.7	75.1	-39.2	76.6	-43.4	78.0	-49.0	79.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.3	17.9	20.5	30.8	42.8	53.0	60.2	66.4	64.9	56.2	43.7	29.9	19.2	
	DBStd	20.72	16.43	15.67	14.29	8.96	8.26	6.04	5.44	6.16	8.34	9.37	13.65	15.07	
	HDD50	4557	995	825	596	243	63	3	0	1	33	234	609	955	
	HDD65	8518	1460	1245	1059	665	378	166	48	78	283	662	1054	1420	
	CDD50	1729	0	0	2	28	157	310	509	464	218	37	4	0	
	CDD65	216	0	0	0	0	7	23	92	76	18	0	0	0	
	CDH74	3720	0	0	1	17	207	395	1388	1263	433	16	0	0	
	CDH80	1274	0	0	0	2	47	97	502	479	146	1	0	0	
Wind	WSAvg	9.8	11.0	10.2	11.1	11.1	10.6	9.2	7.5	7.4	8.5	10.1	10.3	10.7	
Precipitation	PrecAvg	14.3	0.7	0.5	0.7	1.2	1.9	3.3	1.4	1.3	1.2	0.8	0.6	0.6	
	PrecMax	20.9	1.7	1.1	2.0	2.7	4.0	8.1	4.9	3.1	2.7	2.8	1.3	1.3	
	PrecMin	7.6	0.2	0.1	0.0	0.3	0.3	0.6	0.1	0.1	0.0	0.2	0.1	0.0	
	PrecStd	3.0	0.4	0.3	0.5	0.6	0.9	1.6	1.1	0.8	0.8	0.6	0.4	0.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.3	55.5	65.2	76.2	85.5	88.3	93.2	93.0	90.3	76.4	63.8	52.0	
		MCWB	43.2	43.2	48.5	53.0	58.8	65.2	67.3	63.2	60.1	54.8	48.2	41.4	
	2%	DB	45.7	48.3	58.7	70.2	79.9	82.6	89.2	89.2	84.4	69.3	56.6	45.9	
		MCWB	39.0	39.3	45.5	51.0	56.7	62.3	67.1	62.8	59.0	52.5	44.7	38.5	
	5%	DB	41.8	43.6	54.0	64.8	74.7	78.7	85.6	85.5	79.0	63.6	51.5	42.1	
		MCWB	36.4	36.9	43.1	48.0	54.7	60.9	65.7	62.3	57.5	49.4	42.4	36.0	
	10%	DB	38.9	39.6	48.6	59.4	69.9	74.6	82.1	81.6	73.4	58.7	47.0	38.8	
		MCWB	34.7	34.7	39.9	45.3	52.3	58.6	64.7	61.7	55.0	46.8	39.2	33.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.5	44.6	49.5	54.8	61.0	68.4	73.2	68.7	63.5	56.1	50.1	42.8
			MCDB	49.8	54.6	62.9	72.1	78.8	83.1	86.5	84.6	81.7	72.9	61.3	49.9
2%		WB	39.5	40.3	46.5	51.7	58.2	65.0	69.6	66.4	60.5	52.7	45.8	39.1	
		MCDB	45.1	47.7	56.6	68.2	75.5	77.8	83.9	82.0	78.7	66.1	55.4	45.1	
5%		WB	36.7	37.3	43.4	49.1	56.1	62.3	67.5	64.6	58.4	50.0	42.9	36.5	
		MCDB	41.2	42.8	52.9	63.2	72.1	75.3	81.8	79.9	75.5	61.3	51.1	41.7	
10%		WB	34.8	34.9	40.1	46.2	53.9	60.1	65.5	62.7	56.3	47.4	40.0	34.0	
		MCDB	38.9	39.7	47.8	57.5	68.1	72.0	79.3	77.7	72.5	57.5	46.7	38.8	
Mean Daily Temperature Range	5% DB	MDBR	18.8	20.2	20.6	23.1	23.8	22.5	26.9	27.6	26.1	22.7	19.7	18.4	
		MCDBR	19.2	23.2	27.6	33.1	32.3	29.7	33.1	35.5	36.4	30.1	25.1	19.8	
		MCWBR	14.4	15.9	16.4	16.9	14.4	13.9	16.3	15.1	15.8	15.9	15.6	14.1	
	5% WB	MCDBR	19.1	22.6	26.3	30.5	28.5	26.9	30.3	30.1	33.1	28.3	23.8	19.7	
		MCWBR	14.6	16.1	16.3	16.3	13.5	13.8	16.0	14.4	15.1	15.6	15.3	14.4	
Clear-Sky Solar Irradiance	taub		0.265	0.270	0.292	0.341	0.362	0.354	0.354	0.375	0.333	0.292	0.264	0.255	
	taud		2.430	2.476	2.472	2.330	2.317	2.394	2.407	2.319	2.412	2.512	2.498	2.418	
	Ebn at Noon		255	284	294	287	284	287	285	272	274	271	255	242	
	Edh at Noon		19	23	28	36	39	36	35	36	30	23	18	17	
All-Sky Solar Radiation	RadAvg		401	717	1135	1567	1856	2042	2223	1787	1272	793	446	313	
	RadStd		36	58	80	108	135	120	112	113	104	78	30	27	